Lokfix EP

constructive solutions

High strength, thixotropic epoxy resin anchoring grout

Uses

Lokfix EP is ideal to be used in the following operations:

- Installation of starter bars and dowels.
- Base plate bolts.
- Installation of safety barriers and fences.
- For horizontal, vertical and overhead application.

Lokfix EP may also be used as a general-purpose as a fast-setting rapid repair, adhesive and mortar for concrete.

Advantages

- Anchoring without expansion forces.
- Extended working time.
- Good compressive, tensile and flexural strengths.
- High adhesion to substrate.
- Easy to mix and apply.
- Water and corrosion resistant.

Description

Lokfix EP is an epoxy-resin, two component, solvent-free, anchor fixation system for threaded rods and reinforcing bars in uncracked concrete.

When applied it sets and cures to firmly secure a variety of steel fixings into concrete and solid masonry substrates. Lokfix EP may also be used as a general-purpose Structural adhesive for concrete.

Fosroc also offers the Lokfix E product range, cartridge based anchoring products with fully accredited test data and supported by design software. Further information can be found at www.lokfix.com.

Typical Properties

Density	:	1.9 ± 0.1 kg/ltr (at 23°C)
Pot Life	:	45 min at 25˚C
Tack free time	:	5-7 hours at 25˚C
Full Cure	:	7 days at 25˚C
Compressive Strength ASTM C579	:	>60 N/mm^2 (after 7 days)
Flexural strength ASTM C580	:	>28 N/mm^2 (after 7 days)
Tensile strength ASTM D638	:	>40 N/mm^2 (after 7 days)
Elongation ASTM D638	:	>5% (after 7 days)
Bond strength ASTM C882	:	>5 N/mm^2 (after 7 days) (Concrete Failure)
Bond Strength on Steel ASTM D4541	:	>2.7 N/mm^2
Linear Shrinkage ASTM C531	:	<0.08%
Modulus of Elasticity ASTM C580	:	>370 N/mm^2

Table 1: Indictive pull-out values

Steel Diameter	Anchoring Depth	Failure Load
10mm	10cm	>36 KN
12mm	12cm	>39 KN
16mm	16cm	>46 KN
18mm	18cm	>48 KN
20mm	20cm	>50 KN
22mm	22cm	>56 KN
24mm	24cm	>61 KN

Notes on pull-out test values provided on table 1

The results provided are obtained from site pull-out values of deformed steel reinforcement bars in ~40Mpa concrete. All failures occurred in concrete substrate. Testing has been conducted after seven days of anchoring.

Site application procedure was followed as per the TDS noting that the hole was properly cleaned prior to the application of Lokfix EP.

Anchoring depths shown above are tested according to a specific consultant instruction of depth = 10 x diameter.

Site results will vary depending on numerous factors including (but not limited to): - Concrete strength and condition, temperature, Anchoring depth and positioning, loading procedure. Consult with design codes to ensure correct calculation procedure is followed.

In some high strength concrete, failure of steel is the predicted result without any failure in concrete.

Instruction for use

Surface preparation

The drilled hole shall have a diameter at least 4 mm greater than that of the fixing.

The hole shall be thoroughly cleaned using a steel wire bottle brush and an air blower. The brush shall have a diameter at least 1mm wider than the drill bit.

If using diamond drill, the hole must be flushed with clean water, cleaned with a wire brush and flushed again before using the cleaning process described below.

Clean holes immediately prior to installation of fixings to avoid them becoming re-contaminated.

Standing water in the hole shall be removed prior to preparation.

Using a hand pump or compressed air insert the nozzle to the back of the hole and blow out 4 times.

Insert a wire cleaning brush to the bottom of the hole and brush out 4 times.

Using a hand pump or compressed air insert the nozzle to the back of the hole and blow out an additional 4 times.

If dust is still present, repeat the process until no further dust is visible.

Ensure the drill bit and the cleaning brush are of suitable diameter for the fixing used.

Fixings shall be free from rust, paint, grease and contaminants which will interfere with the bond.

Mixing

- The entire contents of the ‘Hardener’ should be poured into the ‘Base’ container and mixed until homogeneous.
- Place the mixed Base and Hardener into a suitable forced action mixer making sure that the entire volume is poured in.
- Once mixed, the material must be used within the specified pot life mentioned in the properties table.

Installation

Installation In Horizontal or Overhead Holes (Lokfix EP):

Use a hollow mastic gun, and a length of tubing to pump the mixed material into the hole, filling from the rear and gradually pulling to the front. Fill it to at least 2/3. Ensure there are no air pockets.

Observing the product gel time, insert the fixing into the hole using a gentle twisting motion.

Ensure the fixing is inserted to the required depth and is held straight until the resin sets.

There should be some extrusion of the Lokfix EP material from the hole which indicates that there is full embedment.

Do not load or apply tension to the fixing until the product cure time has been observed.

Limitation

Minimum substrate and ambient temperature is 4°C.

Do not thin. Addition of solvents will prevent proper cure.

The concrete should be at age 28 days with compressive strength not less than 210 kg/cm^3.

Important Information

The workability, durability and safety of the installed product used for anchoring steel (rebar) depends on the substrate, the dimensions of the element, the drilling and cleaning of holes, the substrate temperature and the type of anchoring bolt or bar.

It is therefore important that a proper structural assessment of the structural elements is carried out by qualified engineers, and that the choice of products, anchor types etc… is based upon such assessment.

Cleaning

Wet resin should be removed from tools and equipment using Fosroc Solvent 102 immediately after use.

Technical Support

Fosroc offers a comprehensive technical support service to specifiers, end users and contractors. It is also able to offer on-site technical assistance, an AutoCAD facility and dedicated specification assistance in locations all over the world.

Estimating

Supply

Lokfix EP	:	0.5 & 2.5 Ltr

Yield

The following calculation of theoretical consumption may be used. Factors such as over-drilling, extrusion from bolt hole, initial gun extrusion and some wastage should also be considered.

Theoretical Consumption:

(π radius cm hole^2 - π radius cm bolt^2) × hole length cm = consumption ml.

Estimated Consumption:

π radius cm hole^2 × hole length cm × 0.67 = consumption ml.

Storage

Shelf life

All products have a shelf life of 12 months if kept in a dry store in the original, unopened bags or packs.

Storage conditions

Store in dry conditions in the original packs. If stored at high temperatures and/or high humidity conditions the shelf life may be reduced.

Important note
Fosroc products are guaranteed against defective materials and manufacture and are sold subject to its standard Conditions for the Supply of Goods and Service. **All Fosroc datasheets are updated on a regular basis. It is the user's responsibility to obtain the recent version.**

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